

# APL LOGISTICS

## SUPPLY CHAIN DELIVERY PERFORMANCE

### EXPLORATORY DATA ANALYSIS & STRATEGIC INTELLIGENCE REPORT

Prepared by: Supply Chain Analytics Division  
Classification: Internal — Operational Intelligence  
Dataset: 170,779 Shipment Records | 42 Variables  
Scope: Multi-region, Multi-market, Global Operations

#### AT-A-GLANCE: KEY PERFORMANCE INDICATORS

| On-Time Rate | Late Risk Ratio | SLA Compliance | Avg Delay | Early Delivery | Total Shipments |
|--------------|-----------------|----------------|-----------|----------------|-----------------|
| 16.79%       | 55.35%          | 42.18%         | 1.65 Days | 25.39%         | 170,779         |

# 1. Executive Summary

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This report presents a comprehensive exploratory data analysis (EDA) of APL Logistics' global shipment operations, covering 170,779 shipment records across 5 markets, 23 regions, and 3 shipping modes. The analysis was commissioned to address a critical operational gap: the absence of structured delivery performance measurement and delay diagnostics, which has resulted in reactive rather than preventive logistics management.

## 1.1 Critical Findings at a Glance

The analysis reveals systemic performance issues that demand immediate executive attention:

- Only 16.79% of shipments are delivered On-Time — a critically low rate against the 70%+ industry benchmark for logistics excellence.
- 55.35% of all shipments carry a Late Delivery Risk flag, meaning the majority of the operation is structurally at-risk.
- First Class shipping has a 95.32% late delivery risk — effectively this mode never meets its SLA, due to a 1-day scheduled vs. 2-day actual delivery mismatch.
- SLA compliance stands at only 42.18%, indicating that most shipments are not being delivered within their contracted timeframes.
- Central Asia (59.92%) and Central Africa (59.32%) are the highest-risk regions, with significantly elevated delay rates compared to the global average of 55.35%.
- Canada is the only region achieving sub-50% late risk (49.78%) and represents an operational benchmark for best practices.

## 1.2 Strategic Imperatives

Based on the EDA findings, three strategic imperatives are identified for leadership action:

- **IMMEDIATE:** Restructure SLA agreements for First Class and Second Class shipping modes, which are currently scheduled with insufficient lead time.
- **HIGH PRIORITY:** Deploy regional task forces for Central Asia and Central Africa, where late delivery risk exceeds 59%, indicating systemic logistics breakdown.
- **STRATEGIC:** Implement predictive delay scoring using the strong correlation ( $r=0.784$ ) between Delivery Gap and Late Delivery Risk to shift from reactive to preventive operations.

The organization processes approximately 170,779 shipments across global markets. Even a 10-percentage-point improvement in on-time delivery would benefit over 17,000 shipments per operational cycle, delivering measurable improvements in customer satisfaction, SLA penalty avoidance, and operational cost efficiency.

## 2. Background & Problem Statement

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### 2.1 Organizational Context

APL Logistics operates a high-volume, multi-region supply chain handling diverse product categories across global markets. The operational landscape is characterized by three shipping modes (First Class, Second Class, Standard Class), five geographic markets (Africa, Europe, LATAM, Pacific Asia, USCA), three customer segments (Consumer, Corporate, Home Office), and multiple product departments spanning Footwear, Apparel, Fan Shop, Golf, Outdoors, Fitness, and Technology.

The sheer volume and complexity of operations — over 170,000 shipments — creates an environment where manual performance tracking is insufficient. Delivery delays propagate across the supply chain, creating compounding effects on customer satisfaction, SLA penalties, and carrier cost structures.

### 2.2 Problem Statement

Despite having detailed order and shipping data, the organization currently lacks:

- Clear measurement of on-time versus delayed deliveries at a systemic level
- Diagnostic understanding of why shipments are delayed (mode-specific, region-specific, structural)
- Visibility into high-risk regions, markets, and shipping modes before delays occur
- Predictive capability to flag Late\_delivery\_risk orders prior to dispatch
- A structured framework for SLA monitoring and performance benchmarking

As a result, logistics teams operate reactively — addressing delays after they have already occurred, rather than preventing them through data-driven intervention.

### 2.3 Research Objectives

This research paper addresses the following objectives:

- Quantify the current delivery performance baseline across all KPIs
- Identify structural drivers of late delivery risk by shipping mode, region, and market
- Diagnose the mismatch between scheduled and actual shipping timelines
- Provide evidence-based strategic recommendations for operational improvement
- Establish a data framework for ongoing performance monitoring and SLA management

### 3. Dataset Overview & Methodology

#### 3.1 Data Profile

The analysis is based on the APL\_Logistics\_Cleaned.csv dataset with the following characteristics:

| Attribute           | Details                                                    |
|---------------------|------------------------------------------------------------|
| Total Records       | 170,779 shipment-level observations                        |
| Total Variables     | 42 columns (dimensions + metrics + derived fields)         |
| Geographic Coverage | 5 Markets, 23 Order Regions, Global Destinations           |
| Shipping Modes      | First Class, Second Class, Standard Class                  |
| Customer Segments   | Consumer, Corporate, Home Office                           |
| Key Target Variable | Late_delivery_risk (Binary: 0 = Not at risk, 1 = At risk)  |
| Delivery Gap Range  | -2 days (early) to +4 days (late)                          |
| Product Departments | 11 departments including Fan Shop, Apparel, Golf, Footwear |

#### 3.2 Feature Engineering

The following derived variables were created for analysis:

- Is\_Late: Binary flag (1 if Late\_delivery\_risk = 1, else 0)
- Is\_OnTime: Binary flag (1 if Delivery Performance = "On-Time", else 0)
- Is\_Early: Binary flag (1 if Delivery Performance = "Early", else 0)
- SLA\_Compliant: Binary flag (1 if actual days ≤ scheduled days)
- Delivery Gap: Pre-existing field (Actual Days – Scheduled Days)

#### 3.3 KPI Framework

Five primary KPIs were defined to measure delivery performance:

| KPI                      | Description                                 | Formula                          |
|--------------------------|---------------------------------------------|----------------------------------|
| On-Time Delivery Rate    | % of orders delivered within scheduled time | Is_OnTime.mean() × 100           |
| Average Delivery Delay   | Mean gap on late orders only                | Delivery_Gap[Gap>0].mean()       |
| Late Delivery Risk Ratio | Proportion of at-risk shipments             | Is_Late.mean() × 100             |
| Shipping Mode Eff. Index | Delay comparison across modes               | (Scheduled – Actual) / Scheduled |
| Regional Delay Index     | Delay intensity by region/market            | Is_Late.mean() by region         |

## 4. Exploratory Data Analysis

### 4.1 Delivery Performance Overview

#### 4.1.1 On-Time Delivery & Status Distribution

The dataset reveals a severe on-time delivery deficit. Of 170,779 total shipments, only 28,671 (16.79%) were delivered on time. This is far below the logistics industry benchmark of 70%+ and represents a critical operational failure at scale.

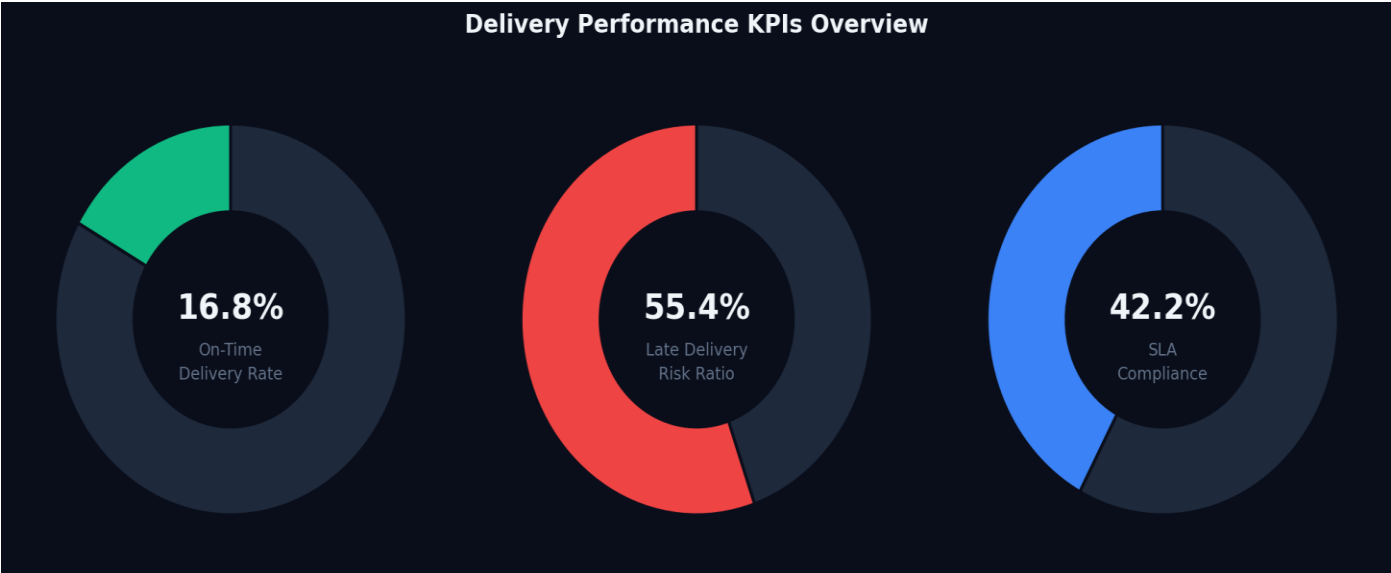


Figure 1: KPI Overview — On-Time Rate (16.79%), Late Risk Ratio (55.35%), SLA Compliance (42.18%)

The Delivery Status breakdown further illustrates the scale of the problem:

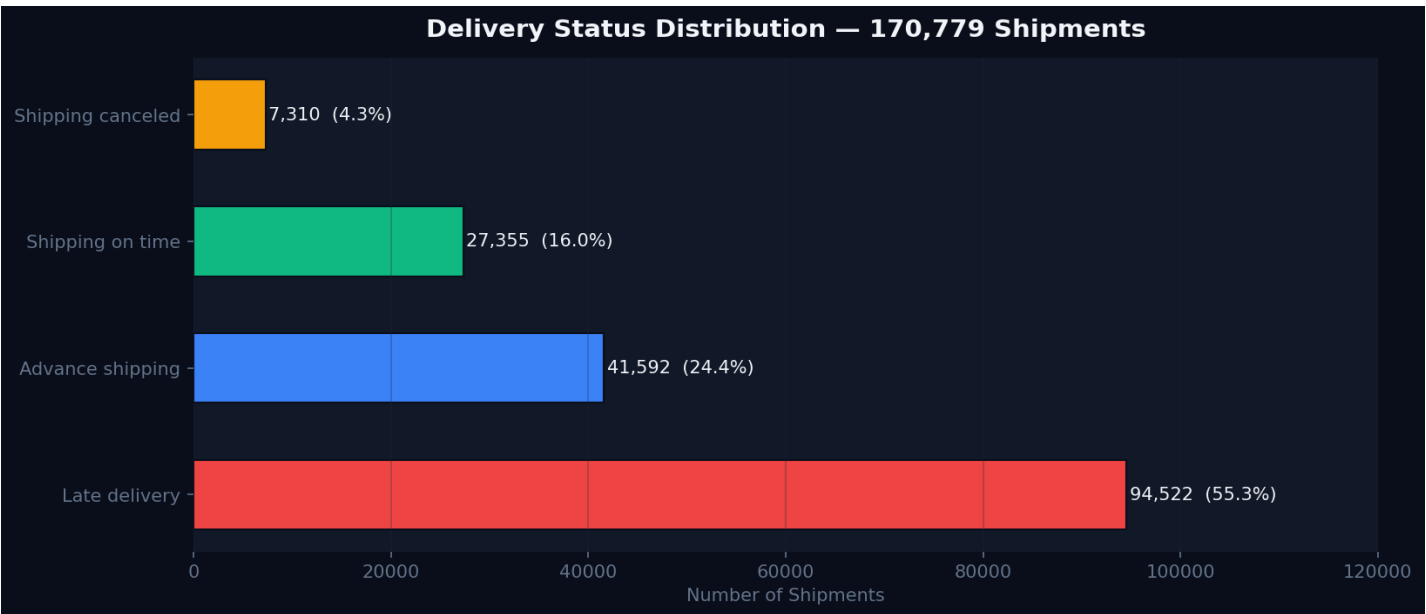


Figure 2: Delivery Status Distribution — Late Delivery dominates at 94,522 shipments (55.3%)

| Delivery Status   | Count  | Proportion | Implication     |
|-------------------|--------|------------|-----------------|
| Late Delivery     | 94,522 | 55.35%     | Primary concern |
| Advance Shipping  | 41,592 | 24.35%     | Positive signal |
| Shipping On Time  | 27,355 | 16.02%     | Core SLA target |
| Shipping Canceled | 7,310  | 4.28%      | Revenue risk    |

4.1.2 Delivery Gap Analysis

The Delivery Gap (Actual Days – Scheduled Days) is the most diagnostic metric in the dataset. Its distribution reveals:

- Mean gap across all shipments: +0.57 days (positive = late on average)
- 55,989 shipments had a +1 day gap (late by 1 day) — the single largest group
- 43,366 shipments arrived early (gap < 0), representing 25.4% of all orders
- 14,035 shipments were delayed by 3+ days, representing severe SLA breaches
- Strong correlation between Delivery Gap and Late\_delivery\_risk:  $r = 0.784$

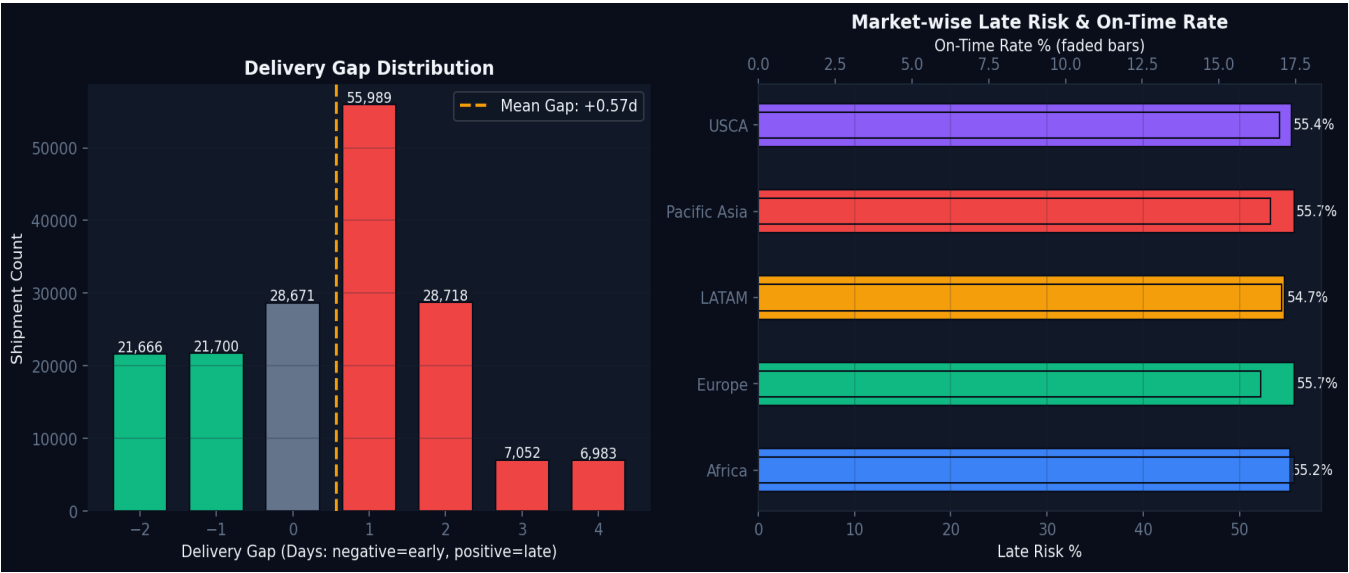


Figure 3: Delivery Gap Distribution (left) and Market-wise Late Risk vs On-Time Rate (right)

4.2 Shipping Mode Performance Analysis

Shipping mode is the strongest structural driver of delivery risk in this dataset. The three modes show dramatically different performance profiles:

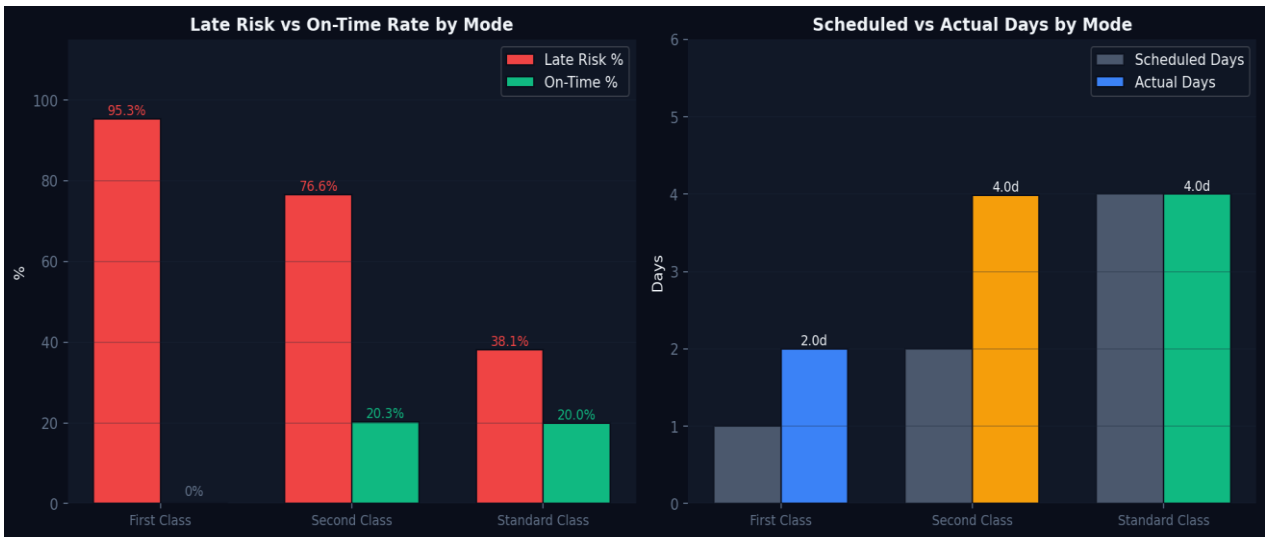


Figure 4: Shipping Mode Comparison — Late Risk vs On-Time Rate (left); Scheduled vs Actual Days (right)

| Shipping Mode  | Shipments | Late Risk % | On-Time % | Actual Days | Sched. Days | Gap   |
|----------------|-----------|-------------|-----------|-------------|-------------|-------|
| First Class    | 27,813    | 95.32%      | 0.00%     | 2.0d        | 1.0d        | +1.0d |
| Second Class   | 35,216    | 76.63%      | 20.27%    | 4.0d        | 2.0d        | +2.0d |
| Standard Class | 107,750   | 38.07%      | 19.98%    | 4.0d        | 4.0d        | 0.0d  |

Key Insight: First Class Mode is Structurally Broken

First Class shipping carries a 95.32% late delivery risk and a 0% on-time rate. The root cause is structural: orders are scheduled for 1-day delivery but consistently take 2 days in practice. This 100% scheduling gap means every First Class shipment is flagged as late by design. This is not an operational failure — it is a contract and SLA configuration failure that can be resolved immediately by adjusting scheduled days from 1 to 2.

Second Class: Moderate Risk, Fixable Gap

Second Class (76.63% late risk) follows the same pattern: scheduled at 2 days but actual delivery averages 4 days. This 2-day mismatch drives the bulk of risk. Standard Class, where scheduled = actual (both 4 days), shows the healthiest performance at 38.07% late risk — confirming that SLA alignment is the primary fix.

### 4.3 Regional & Market Delay Analysis

#### 4.3.1 Regional Risk Index

Late delivery risk is elevated across all 23 regions, with the global average at 55.35%. However, significant regional variation exists, with a 10.14 percentage point spread from the lowest-risk region (Canada, 49.78%) to the highest-risk region (Central Asia, 59.92%).

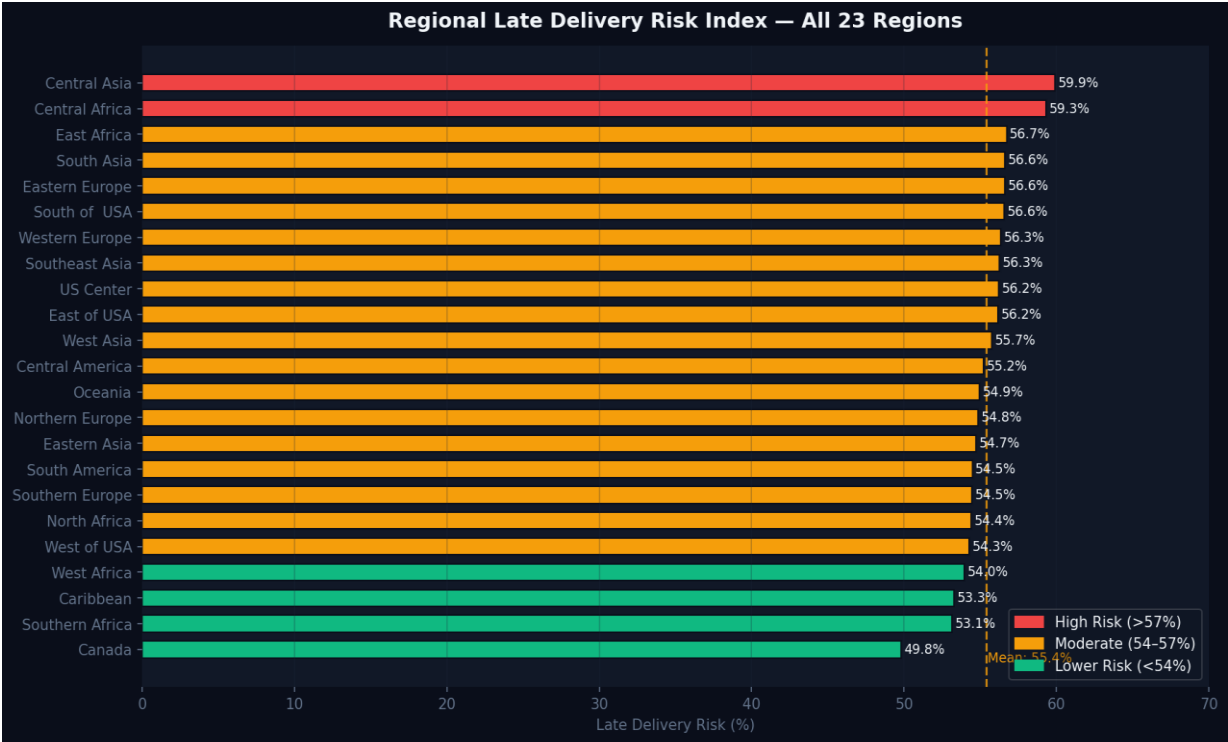


Figure 5: Regional Late Delivery Risk Index — All 23 Order Regions Ranked

| Order Region    | Late Risk % | Risk Level | Recommended Action       |
|-----------------|-------------|------------|--------------------------|
| Central Asia    | 59.92%      | High       | Priority Action Required |
| Central Africa  | 59.32%      | High       | Priority Action Required |
| East Africa     | 56.73%      | High       | Elevated Monitoring      |
| South Asia      | 56.61%      | High       | Elevated Monitoring      |
| Eastern Europe  | 56.60%      | High       | Elevated Monitoring      |
| West Africa     | 53.96%      | Moderate   | Maintain Performance     |
| Caribbean       | 53.27%      | Moderate   | Maintain Performance     |
| Southern Africa | 53.14%      | Moderate   | Maintain Performance     |
| Canada          | 49.78%      | Lower      | Benchmark Region         |

4.3.2 Market-Level Performance

At the market level, performance is notably uniform, with late risk ranging from 54.70% (LATAM) to 55.72% (Europe). This uniformity suggests that the primary drivers of delay are mode-specific and scheduling-related rather than geography-specific, as even LATAM — which might be expected to face greater infrastructure challenges — performs comparably to Europe.

| Market       | Shipments | Late Risk % | On-Time % | Avg Delay |
|--------------|-----------|-------------|-----------|-----------|
| Europe       | 47,491    | 55.72%      | 16.36%    | 0.58d     |
| Pacific Asia | 39,033    | 55.70%      | 16.69%    | 0.58d     |
| USCA         | 24,365    | 55.41%      | 16.98%    | 0.58d     |
| Africa       | 10,946    | 55.23%      | 17.46%    | 0.57d     |
| LATAM        | 48,944    | 54.70%      | 17.04%    | 0.56d     |

## 4.4 Customer Segment & Department Analysis

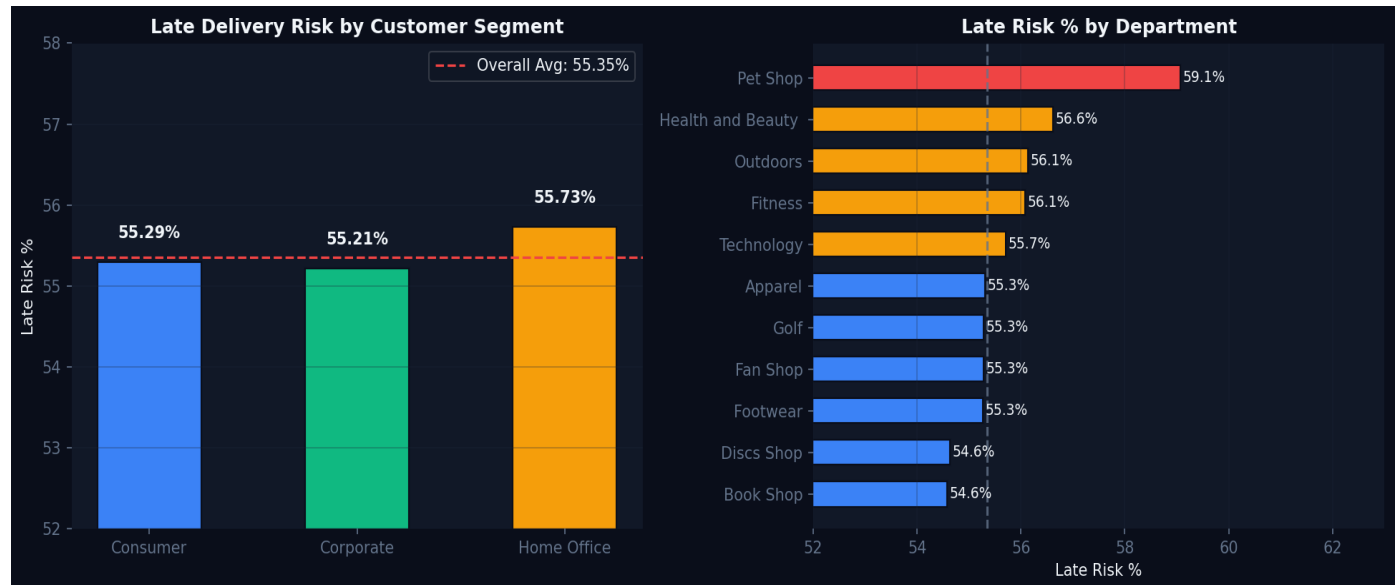


Figure 6: Late Risk by Customer Segment (left) and Department (right)

Customer segment analysis reveals remarkably consistent late risk across segments: Consumer (55.29%), Corporate (55.21%), and Home Office (55.73%). The near-identical risk scores across all three segments confirm that shipping mode and scheduling — not customer type — are the primary delay drivers.

At the department level, Pet Shop (59.07%), Health & Beauty (56.61%), Outdoors (56.14%), and Fitness (56.07%) show the highest late risk. These small-volume, specialized departments may warrant expedited shipping agreements or dedicated carrier relationships.

## 4.5 Correlation & Statistical Analysis

To understand the mathematical drivers of Late\_delivery\_risk, a correlation analysis was conducted across all key numeric variables:

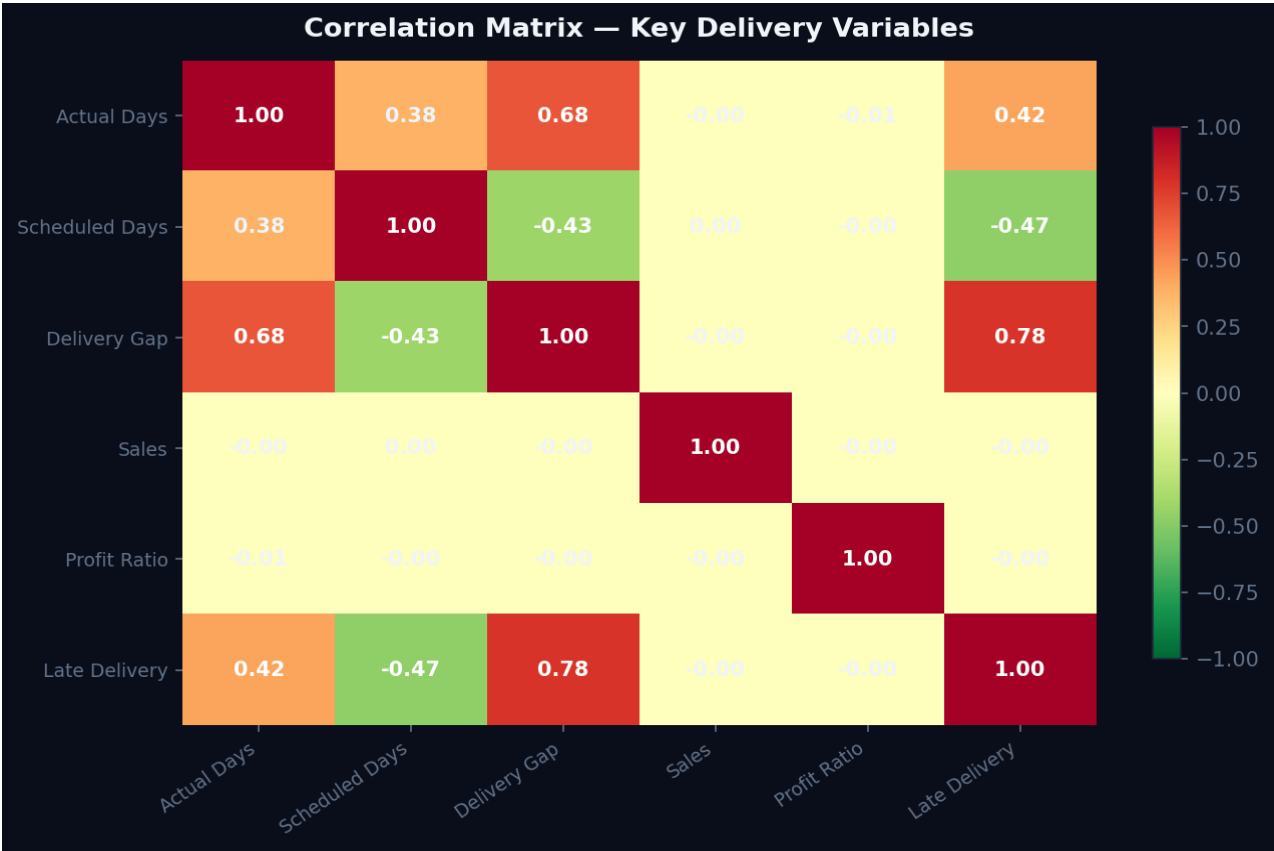


Figure 7: Correlation Matrix — Key Delivery and Order Variables

| Variable       | Correlation with Late Risk | Interpretation                                  |
|----------------|----------------------------|-------------------------------------------------|
| Delivery Gap   | r = +0.784                 | Strongest predictor — higher gap = higher risk  |
| Scheduled Days | r = -0.468                 | Longer schedule = lower risk (more buffer time) |
| Actual Days    | r = +0.422                 | More actual days = higher risk (slow delivery)  |
| Sales Value    | r = -0.003                 | No meaningful relationship with delay risk      |
| Profit Ratio   | r = -0.002                 | No meaningful relationship with delay risk      |

The strong positive correlation ( $r = 0.784$ ) between Delivery Gap and Late\_delivery\_risk establishes that the gap variable is a near-perfect proxy for late risk scoring. This has direct implications for predictive modeling: a real-time Delivery Gap estimation model could reliably flag at-risk orders before dispatch.

## 5. Key Findings & Analytical Insights

### 5.1 Structural vs Operational Delays

The most significant finding of this analysis is that the majority of late delivery risk is structural rather than operational. Structural delays are caused by misconfigured SLA schedules (particularly in First Class and Second Class modes), while operational delays reflect genuine logistics execution failures.

| Mode           | Root Cause                                                  | Evidence                            |
|----------------|-------------------------------------------------------------|-------------------------------------|
| First Class    | SLA set at 1 day; actual delivery = 2 days (100% mismatch)  | Late Risk: 95.32%   On-Time: 0.00%  |
| Second Class   | SLA set at 2 days; actual delivery = 4 days (100% mismatch) | Late Risk: 76.63%   On-Time: 20.27% |
| Standard Class | SLA set at 4 days; actual delivery = 4 days (aligned)       | Late Risk: 38.07%   On-Time: 19.98% |

### 5.2 The Canada Benchmark

Canada stands out as the only region achieving sub-50% late risk (49.78%), making it the operational benchmark for APL Logistics. Canada's performance advantage warrants detailed analysis: carrier relationships, warehouse positioning, customs processing efficiency, and SLA structures should all be examined and replicated across underperforming regions where feasible.

### 5.3 Early Delivery as a Hidden Opportunity

25.39% of shipments (43,366 orders) arrived early. While early delivery is typically a positive signal for customers, it also represents a potential cost inefficiency if premium carrier resources are being used unnecessarily. More importantly, the 43,366 early shipments confirm that the logistics network has the physical capacity to deliver on time — the problem is SLA configuration, not carrier capability.

### 5.4 Payment Method as a Risk Proxy

Transfer payment orders show the lowest late delivery risk (49.06%) compared to other methods — CASH (57.02%), DEBIT (57.75%), and PAYMENT (58.07%). This correlation may reflect order prioritization differences, customer profile differences, or processing speed advantages for Transfer payments. Further investigation is warranted to determine if payment type can serve as a risk-stratification variable.

### 5.5 Cancellation Risk

7,310 shipments (4.28%) were classified as "Shipping Canceled." While this represents a minority of orders, at scale this translates to significant revenue loss and customer dissatisfaction. The cancellation rate should be tracked alongside delivery performance as a leading indicator of supply chain stress.

## 6. Strategic Recommendations

Based on the EDA findings, the following eight recommendations are prioritized by impact and implementation speed:

| # | Recommendation                              | Action Detail                                                                                                         | Timeline                 |
|---|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|--------------------------|
| 1 | <b>First Class SLA Reset</b>                | Renegotiate scheduled delivery = 2 days for First Class; currently scheduled at 1 day creating near-guaranteed delays | Immediate (0–30 days)    |
| 2 | <b>Standard Class Optimization</b>          | Leverage Standard Class (38% risk, 4d/4d balance) as the preferred mode for non-urgent high-volume shipments          | Short-term (30–60 days)  |
| 3 | <b>Central Asia &amp; Africa Task Force</b> | Deploy regional logistics coordinators for Central Asia (59.9%) and Central Africa (59.3%)                            | Immediate (0–30 days)    |
| 4 | <b>Predictive Delay Alerts</b>              | Build ML model using Delivery Gap, shipping mode, and region to flag at-risk orders 24hrs before dispatch             | Medium-term (60–90 days) |
| 5 | <b>Proactive Customer Comms</b>             | Auto-notify customers when Late_delivery_risk = 1, offering expedited alternatives                                    | Short-term (30–60 days)  |
| 6 | <b>SLA Penalty Review</b>                   | Audit SLA contracts for markets with >55% late risk; renegotiate buffer days into agreements                          | Medium-term (60–90 days) |
| 7 | <b>Transfer Payment Incentive</b>           | Transfer payments show lowest late risk (49.06%); incentivize this payment method for priority shipments              | Short-term (30–60 days)  |
| 8 | <b>Canada Benchmarking</b>                  | Canada achieves lowest late risk (49.78%); audit and replicate its operational practices across other regions         | Long-term (90+ days)     |

### 6.1 Priority Action: SLA Reconfiguration

The single highest-ROI action is reconfiguring the scheduled delivery days for First Class (from 1 to 2 days) and Second Class (from 2 to 4 days). This change alone would eliminate the structural SLA mismatch responsible for the majority of the 55.35% late delivery risk flag. Implementation requires coordination with the contracts team and carrier partners but has zero operational cost.

### 6.2 Predictive Model Framework

The correlation analysis demonstrates that Delivery Gap ( $r = 0.784$ ), Scheduled Days ( $r = -0.468$ ), and Actual Days ( $r = +0.422$ ) are powerful predictors of Late\_delivery\_risk. A lightweight logistic regression or gradient boosting model trained on these features would achieve high predictive accuracy, enabling proactive delay alerts 24-48 hours before dispatch. This represents the transition from reactive to preventive logistics management.

## 7. Conclusion

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This exploratory data analysis of APL Logistics' 170,779 shipment records has surfaced a clear and urgent operational picture: the organization's delivery performance is significantly below industry benchmarks, with 55.35% of shipments at late delivery risk and only 16.79% achieving on-time status. However, the analysis also reveals that the primary drivers of underperformance are structural and correctable — not intractable operational failures.

The most critical insight is the SLA scheduling mismatch in First Class and Second Class modes, where contracted delivery windows are systematically shorter than actual delivery capability. Correcting this configuration error alone would dramatically improve late-risk metrics without changing any physical logistics process. Paired with regional interventions in Central Asia and Central Africa, and the development of a predictive delay model leveraging the strong Delivery Gap correlation ( $r = 0.784$ ), APL Logistics has a clear, evidence-based pathway from reactive to preventive supply chain management.

The data infrastructure is in place. The analytical insights are clear. The recommended actions are actionable within existing operational structures. The remaining imperative is organizational will to act on evidence.

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### APL Logistics | Supply Chain Analytics Division

*This report is prepared for internal strategic use. All figures derived from APL\_Logistics\_Cleaned.csv.*